Course TA F0

CST3492 Java Enterprise Application Development

Assignment A | Group B | Release 0.1.0

Full names and matriculation numbers

Name _____ ID _____

Name _____ ID _____

Name _____ ID _____

Name _____ ID _____

F0 functionality

Course questions with evidence

Local lecture retrieval supplies relevant passages to a configurable LLM.

Document and page references

Course answers cite source IDs that resolve to the original document and page.

Questions about selected Java code

An editor action attaches the selected text and line range to the tool window.

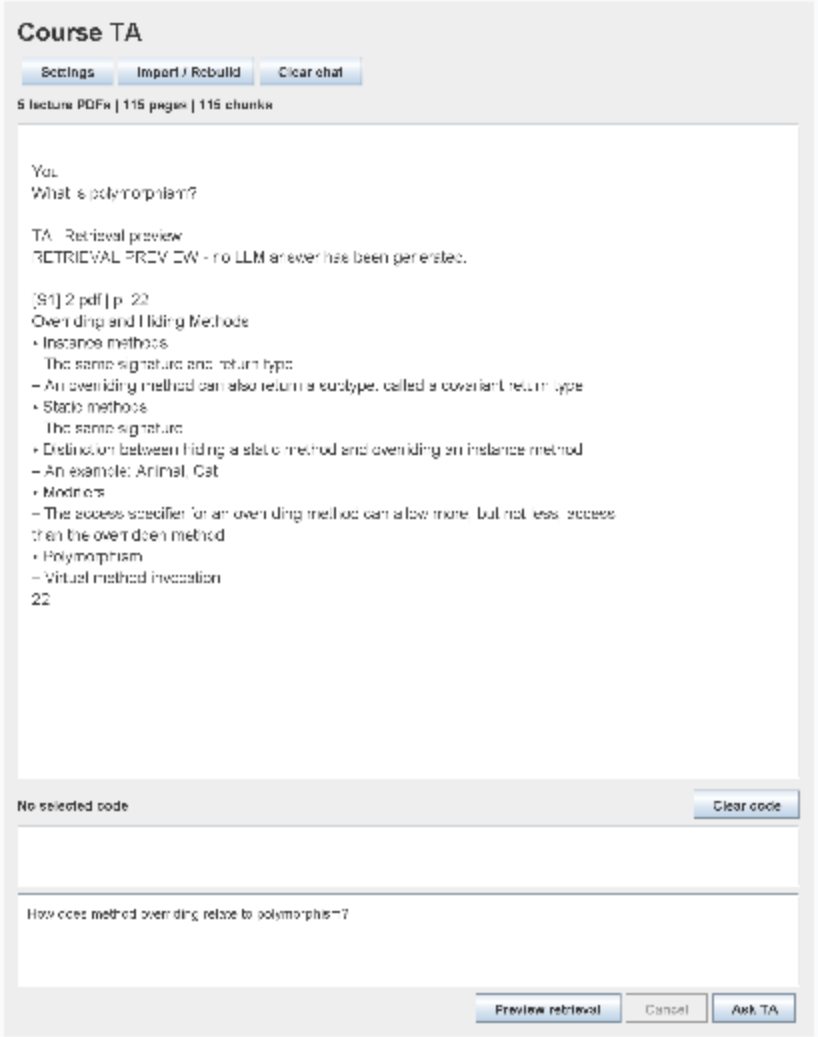
Course material retrieval

Import the five lecture PDFs.

Preview a question before configuring a model.

Inspect the source document and physical page number.

Example
Polymorphism → 2.pdf, page 22



Offscreen UI sample; no LLM call

Selected code context

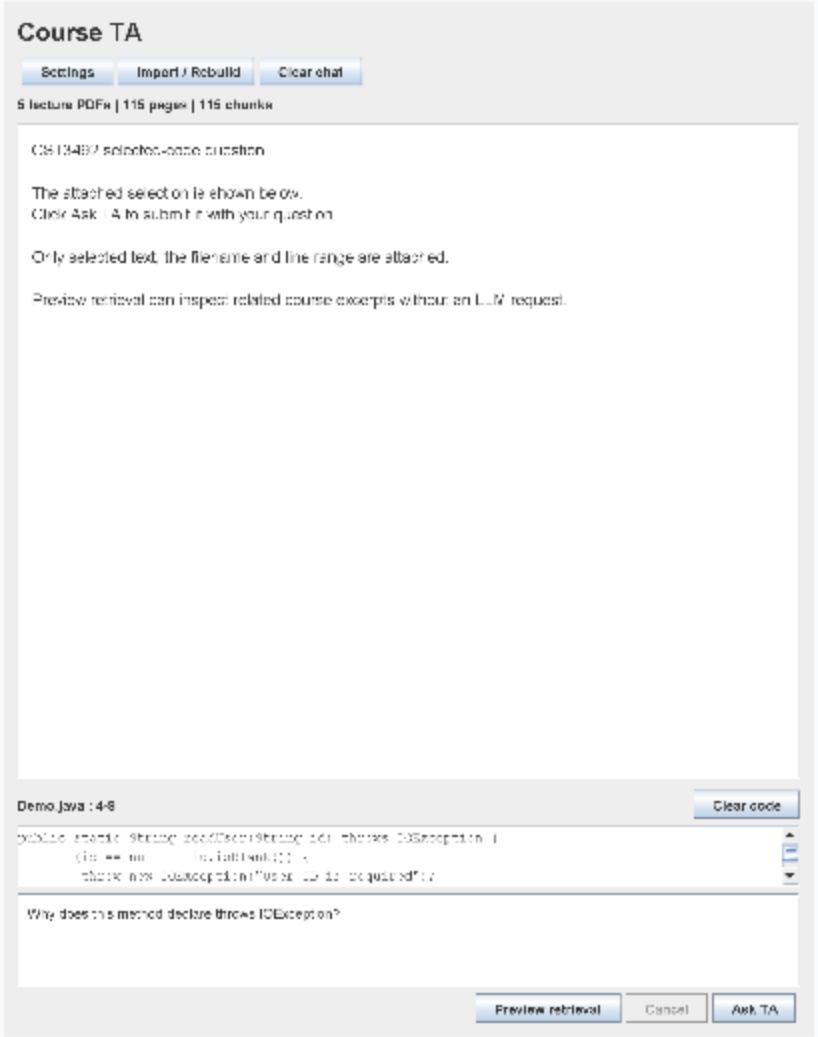
1 Select code in the editor.

2 Right-click Ask Course TA About Selected Code.

3 Enter a question and review the attachment.

4 Click Ask TA.

The rest of the project is not sent.



Offscreen UI component sample

Architecture

IDE layer

Tool window, editor action, settings and background task control.

Local RAG layer

PDFBox extraction → page-preserving chunks → BM25 index → source IDs.

LLM layer

Question + selected code + retrieved excerpts → HTTP request → citation checks.

Chunking and retrieval

1,400 characters per chunk
200-character target overlap
No chunk crosses a PDF page

BM25 ranks local passages using an inverted index.

The relevance gate requires score ≥ 1.0 and ≥ 20% query-term coverage.

At most four excerpts are included in each model request.

Citation and fallback behavior

Relevant course evidence

The prompt assigns [S1], [S2] and other IDs. The answer must cite valid IDs.

No supporting course evidence

The answer explicitly states that it is based on general knowledge.

Missing or invalid references

The application withholds the answer and asks for a retry. Semantic accuracy still requires review.

Validation results

5 lecture PDFs | 115 pages | 115 chunks

18 automated tests passed

PDF provenance, retrieval, persistence, prompt limits, citation gates and HTTP handling are covered.

Plugin ZIP built against IntelliJ IDEA 2026.1.1.

Live provider and interactive IDE demonstration remain to be checked by the team.

Team demonstration and F1 milestone

F0 demonstration

Course question and page citation

Selected-code question

General-knowledge fallback

Architecture and design limits

Group B next milestone

Scenario-based JUnit tests

Staged-diff commit messages

One creative smart action

Actual team roles and collaboration evidence: _____